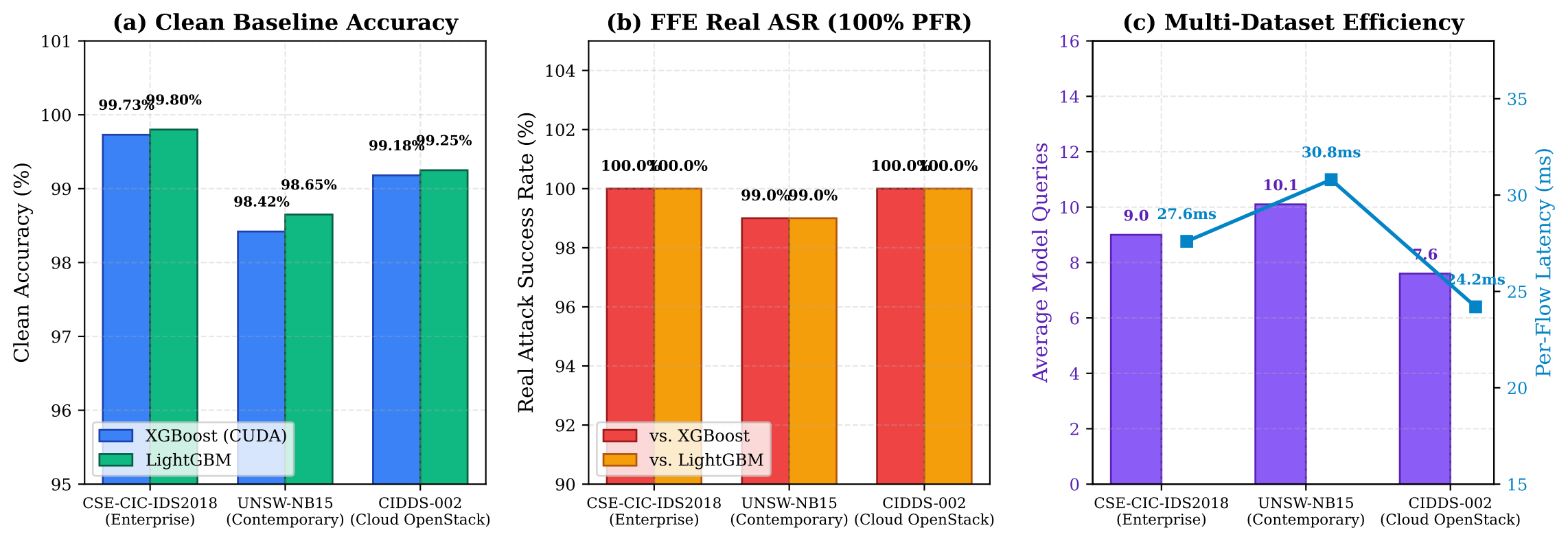
(a) Clean Baseline Accuracy
Clean Accuracy (%)
101
100
99
98
97
96
95
99.73%
99.80%
98.42%
98.65%
99.18%
99.25%
XGBoost (CUDA)
LightGBM
CSE-CIC-IDS2018 (Enterprise)
UNSW-NB15 (Contemporary)
CIDDS-002 (Cloud OpenStack)
(b) FFE Real ASR (100% PFR)
Real Attack Success Rate (%)
104
102
100
98
96
94
92
90
100.0%
100.0%
99.0%
99.0%
100.0%
100.0%
vs. XGBoost
vs. LightGBM
CSE-CIC-IDS2018 (Enterprise)
UNSW-NB15 (Contemporary)
CIDDS-002 (Cloud OpenStack)
(c) Multi-Dataset Efficiency
Average Model Queries
Per-Flow Latency (ms)
16
14
12
10
8
6
4
2
0
35
30
25
20
15
9.0
27.6ms
10.1
30.8ms
7.6
24.2ms
CSE-CIC-IDS2018 (Enterprise)
UNSW-NB15 (Contemporary)
CIDDS-002 (Cloud OpenStack)